

Groups of automorphisms on Hilbert spaces and probability algebras

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Groups of automorphisms

Recall that *metric structures* $\mathcal{M}$ are constructed over complete, bounded metric spaces.

Fix a countable group G .

Notation

Fix $\mathcal{L}$ a countable, metric, first order language and let T be a consistent $\mathcal{L}$ -theory.

We denote

$$\mathcal{L}^G = \mathcal{L} \cup \{\tau_g \mid g \in G\}$$

$$T_G = T \cup \{“\tau_g \text{ is an automorphism”} \mid g \in G\}$$

Models of T_G are of the form $(\mathcal{M}, \tau_g)_{g \in G}$, with $\mathcal{M} \models T$ and $(\tau_g)_{g \in G}$ an action via $\mathcal{L}$ -automorphisms.

We are interested in finding the model companion T_G^+ of T_G .

Hilbert spaces

Let $\mathcal{H}$ be a Hilbert space, let $B_{\mathcal{H}}$ be its closed unit ball. We may identify $\mathcal{H}$ with the structure

$$\mathcal{H} = \left(B_{\mathcal{H}} ; \frac{x+y}{2} , - , 0 , (\lambda_z)_{z \in \mathbb{Q}(i), |z| \leq 1} , \langle ; \rangle , \| \cdot \| , d \right).$$

Let $\mathcal{L}_{\mathbf{H}}$ be the language and **HS** their common theory.

The model companion of **HS** is the theory of infinite-dimensional Hilbert spaces **IHS**.

Fact ([YBHU08])

IHS is κ -categorical all infinite κ , so it is complete and ω -stable.
It admits quantifier elimination in $\mathcal{L}_{\mathbf{H}}$.

Models $(\mathcal{H}, \pi_g)_{g \in G}$ of **HS**_G are exactly unitary representations of G . We write $(\mathcal{H}, \pi)$ instead.

Probability algebras

Let $(X, \mathcal{A}, \mu)$ be a probability space, $\mathfrak{N}_\mu = \{A \subseteq X \mid \mu(A) = 0\}$. Then $\mathbb{A} = \mathcal{A}/\mathfrak{N}_\mu$ is complete measure-algebra, with σ -additive measure μ and a complete metric $d(a, b) = \mu(a \triangle b)$. We may regard $\mathbb{A}$ as the metric structure

$$\mathbb{A} = \left(\mathbb{A} ; \triangle , \cup , \cap , \quad ^c , \mathbb{0} , \mathbb{1} , \mu , d \right)$$

where $\mathbb{0} = \widetilde{\emptyset}$ and $\mathbb{1} = \widetilde{X}$.

Let $\mathcal{L}_{\mathbf{Pr}}$ be the language and $\mathbf{Pr}$ their common theory.

An *atom* is an event $a \in \mathbb{A}$ with $a \neq \mathbb{0}$ such that $b \subseteq a$ implies $b = a$ or $b = \mathbb{0}$.

The model companion of $\mathbf{Pr}$ is the theory of atomless probability algebras $\mathbf{APr}$.

Fact ([BH23])

$\mathbf{APr}$ is ω -categorical and thus complete. The only separable model is the one associated to $([0, 1], \mathfrak{L}, \lambda)$. It is not $\aleph_1$ -categorical, but Maharam's Theorem classifies its models.

It admits quantifier elimination in $\mathcal{L}_{\mathbf{Pr}}$ and it is ω -stable.

Probability algebras

Suppose G acts on $(X, \mathcal{A}, \mu)$ and suppose $x \rightarrow g \cdot x$ is measurable and *measure-preserving*, meaning that $\mu(A) = \mu(g \cdot A)$ for all measurable $A \subseteq X$. The action induces $(\mathbb{A}, \tau_g)_{g \in G} \models \mathbf{Pr}_G$ by defining:

$$\tau_g(\tilde{A}) = \widetilde{(g \cdot A)}, \quad \tilde{A} \in \mathbb{A}.$$

Fact (Folklore)

All models of $\mathbf{Pr}_G$ are of this form.

The action is *free* if $\mu(x \in X \mid g \cdot x = x) = 0$ for $g \neq 1_G$ in G .

They usually correspond to existentially closed actions (*a priori*, not clear how to axiomatize them).

Unitary representations of a finite group

Fix a finite group G .

For a family $(\mathcal{H}_\alpha, \pi_\alpha)_\alpha$ of unitary representations, we can form its direct sum $\bigoplus_\alpha (\mathcal{H}_\alpha, \pi_\alpha)$. We write $\kappa(\mathcal{H}, \pi)$ for $\bigoplus_{\alpha < \kappa} (\mathcal{H}, \pi)$.

Fact

For a finite group G , there are finitely many irreducible finite-dimensional representations $W_1, W_2, \dots, W_k$ such that any model $(\mathcal{H}, \pi)$ is of the form

$$(\mathcal{H}, \pi) \cong \kappa_1 W_1 \oplus \kappa_2 W_2 \oplus \dots \oplus \kappa_k W_k.$$

For example, let G act on itself by multiplication on the left. Consider the induced representation $(\ell_2(G), \pi)$, called the *regular representation* of G . Then

$$\ell_2(G) \cong d_1 W_1 \oplus d_2 W_2 \oplus \dots \oplus d_k W_k$$

with $d_i = \dim W_i$.

Unitary representations of a finite group

Let $(\mathcal{H}, \pi) = \kappa_1 W_1 \oplus \kappa_2 W_2 \oplus \cdots \oplus \kappa_k W_k \models \mathbf{HS}_G$. We find an axiomatization $\mathbf{HS}_\pi$ of $\text{Th}(\mathcal{H}, \pi)$.

Let $P_i : \mathcal{H} \longrightarrow \kappa_i W_i$ be the orthogonal projection. Then

$$P_i x = \frac{\dim W_i}{|G|} \sum_{g \in G} \overline{\chi_{W_i}(g)} \pi_g(x),$$

is a definable function, and thus $\kappa_i W_i$ is definable.

$\mathbf{HS}_\pi$ needs to specify all values of $\dim(\kappa_i W_i)$. It is ω -categorical and thus complete.

Theorem (Berenstein, Pérez. [BP24])

$\mathbf{HS}_\pi$ admits quantifier elimination in $\mathcal{L}_H^G$ and it is ω -stable.

It follows that the model-companion $\mathbf{IHS}_G^+$ is the theory with all κ_i infinite. This is an existential axiomatization. Its only separable model is

$$\infty \ell_2(G) = \infty W_1 \oplus \infty W_2 \oplus \cdots \oplus \infty W_k.$$

Measure-preserving actions of a finite group

Let G act on $(X, \mathcal{A}, \mu)$.

Fact (Rokhlin's lemma for finite groups; [Hal56])

Assume G is finite. Then a measure-preserving action of G on an atomless probability space $(X, \mathcal{A}, \mu)$ is free iff there is $A \subseteq X$ such that $(g \cdot A \mid g \in G)$ is a disjoint, measurable partition of X .

Thus we can give existential axiomatization $\mathbf{APr}_G^+$ for free actions.

Corollary

Two free measure-preserving actions on $([0, 1], \mathfrak{L}, \lambda)$ are isomorphic.

Let $\mathbf{APr}_G^+$ be the theory of free actions of G , axiomatized by Rokhlin's lemma. Then $\mathbf{APr}_G^+$ is ω -categorical and thus complete.

Furthermore, $\mathbf{APr}_G^+$ admits quantifier elimination and it is ω -stable.

Unitary representations of an amenable group

If G is amenable, there is an existential axiomatization if $\mathbf{IHS}_G^+$ given in [Ber07]. The abelian case is easier.

Example ($G = \mathbb{Z}$)

For $\pi \in \mathcal{U}(\mathcal{H})$, then $(\mathcal{H}, \pi)$ can be seen as a model of $\mathbf{HS}_{\mathbb{Z}}$ defining $\pi_n = \pi^n$. Suppose $(\mathcal{H}, \pi) \models \mathbf{IHS}_{\mathbb{Z}}^+$.

As π is unitary, then $\sigma(\pi) \subseteq S^1$. It is existentially closed iff $\sigma(\pi) = S^1$. The forward direction holds as

$$\lambda \in \sigma(\pi) \quad \text{iff} \quad (\mathcal{H}, \pi) \models \inf_{\|x\|=1} \|\pi x - \lambda x\| = 0.$$

For $\lambda \in S^1$, there is a type $p_{\lambda}(x)$ containing “ $\|x\|=1$ ” and “ $\pi x = \lambda x$ ”. This type can be omitted or realized in a separable model, so $\mathbf{IHS}_{\mathbb{Z}}^+$ is not ω -categorical.

If $\lambda \neq \lambda'$, fix $x \models p_{\lambda}$ and $y \models p_{\lambda'}$. Then $x \perp y$ and so $\|x - y\| = \sqrt{2}$. Thus $d(p_{\lambda}, p_{\lambda'}) = \sqrt{2}$, so $\mathbf{IHS}_{\mathbb{Z}}^+$ is not ω -stable.

Unitary representations of an amenable group

The abelian case is easier.

Example ($G = \mathbb{Z}^2$)

If $\pi, \pi' \in \mathcal{U}(\mathcal{H})$ commute, then $(\mathcal{H}, \pi, \pi')$ can be seen as a model of $\mathbf{HS}_{\mathbb{Z}^2}$. Define

$$\sigma(\pi, \pi') = \left\{ (\lambda, \lambda') \mid (\mathcal{H}, \pi, \pi') \models \inf_{\|x\|=1} \max(\|\pi(x) - \lambda x\|, \|\pi'(x) - \lambda' x\|) = 0 \right\}.$$

Then $(\mathcal{H}, \pi, \pi')$ is existentially closed iff $\sigma(\pi, \pi') = S^1 \times S^1$

Example ($G = (\mathbb{Z}/4\mathbb{Z}) \times \mathbb{Z}$)

Let $\pi, \pi' \in \mathcal{U}(\mathcal{H})$ commute with $\pi^4 = I_{\mathcal{H}}$, so $(\mathcal{H}, \pi, \pi') \models \mathbf{IHS}_G$.

It is existentially closed iff $\sigma(\pi, \pi') = \{1, i, -1, -i\} \times S^1$.

Something similar can be done whenever G is abelian, by the structure theorem for finitely generated abelian groups.

Measure-preserving actions of an amenable group

If G is amenable, there is a version of Rokhlin's Lemma for free actions. Thus we can axiomatize a theory $\mathbf{APr}_G^+$ of free actions. It is the model companion of $\mathbf{Pr}_G$.

Fact (Rokhlin's lemma, [Hal56])

Let τ be an invertible, measure-preserving transformation on an atomless probability space $(X, \mathcal{A}, \mu)$. The induced $\mathbb{Z}$ -action is free iff, for $n > 1$ and $\varepsilon > 0$ there is $A \subseteq X$ such that $\mu\left(\bigcup_{k \leq n} \tau^k(A)\right) > 1 - \varepsilon$.

Fix a Følner sequence $(F_n)_n$ for G .

Fact (Rokhlin's lemma for amenable groups. [OW87])

A measure-preserving action of G on an atomless probability space $(X, \mathcal{A}, \mu)$ is free iff, for $n > 1$ and $\varepsilon > 0$ there is $A \subseteq X$ such that $\mu\left(\bigcup_{g \in F_n} g \cdot A\right) > 1 - \varepsilon$.

Measure-preserving actions of an amenable group

It follows from Rokhlin's lemma that two free, measure-preserving actions on $([0, 1], \mathcal{L}, \mu)$ are *approximately isomorphic* in the following sense:

Theorem

Let $(\mathbb{A}, \tau_g)_{g \in G}, (\mathbb{A}, \tau'_g)_{g \in G} \models \mathbf{APr}_G^+$ be the induced models. For all finite $K \subseteq G$ and $\varepsilon > 0$ there is $\rho \in \text{Aut}(\mathbb{A})$ with $\sup_{a \in \mathbb{A}} d(\rho \tau_g \rho^{-1}(a), \tau'_g(a)) < \varepsilon$ for $g \in K$.

Then, they must have isomorphic ultrapowers. Thus $\mathbf{APr}_G^+$ is complete. The theory is not ω -categorical:

Example (Bernoulli shifts)

Let $(K, \mathcal{B}, \nu)$ be a finite probability space. Let G act on $(X, \mathcal{A}, \mu) = (K, \mathcal{B}, \nu)^G$ by permuting coordinates, and let $(\mathbb{A}_\nu, \tau_g)_{g \in G} \models \mathbf{APr}_G^+$ be the corresponding model.

The *entropy* of ν is $H(\nu) = - \sum_{k \in K} \nu(k) \log \nu(k)$.

Ornstein and Weiss proved that, $(\mathbb{A}_\nu, \tau_g)_{g \in G} \cong (\mathbb{A}_{\nu'}, \tau_g)_{g \in G}$ iff $H(\nu) = H(\nu')$.

Thus $\mathbf{APr}_G^+$ has $2^{\aleph_0}$ different separable models.

Measure-preserving actions of amenable groups

Theorem (Berenstein, Henson, S.)

The theory $\mathbf{APr}_G^+$ admits quantifier elimination in $\mathcal{L}_{Pr}^G$ and thus it is stable. Non-forking can be characterized by conditional probabilities.

The theory $\mathbf{APr}_G^+$ need not be ω -stable.

Example (Irrational rotations)

For irrational θ , let $\mathbb{Z}$ act on $([0, 1], \mathfrak{L}, \lambda)$ by $\tau_\theta(x) = x + n\theta \pmod{1}$, and let $(\mathbb{A}, \tau_\theta) \models \mathbf{APr}_{\mathbb{Z}}^+$ be the associated model.

Then $(\mathbb{A}, \tau_\theta) \cong (\mathbb{A}, \tau_{\theta'})$ iff $\theta = \theta'$ or $\theta = -\theta' \pmod{1}$.

Furthermore, let $p_\theta(x)$ be the type of $\widetilde{[0, \frac{1}{2}]}$ in $(\mathbb{A}, \tau_\theta)$. If $\theta, \theta', \theta/\theta'$ are irrational, then $d(p_\theta, p_{\theta'}) \geq 1/2$, so $\mathbf{APr}_{\mathbb{Z}}^+$ is not ω -stable.

The argument can be adapted whenever G is an finitely generated, infinite abelian group. We don't know for any infinite group.

The Koopman representation

If G acts on $(X, \mathcal{A}, \mu)$ and $\mathbb{A} = \mathcal{A}/\mathfrak{N}_\mu$, define $(L_2(\mathbb{A}), \pi_g)_{g \in G}$ by

$$\pi_g f(x) = f(g \cdot x)$$

Thus $(L_2(\mathbb{A}), \pi)$ is determined by $(\mathbb{A}, \tau_g)_{g \in G}$ as $\pi_g \chi_a = \chi_{\tau_g(a)}$. This is the *Koopman representation* of $(\mathbb{A}, \tau_g)_{g \in G}$.

Theorem (Bernstein, S.)

Assume G is amenable and residually finite.

If $(\mathbb{A}, \tau_g)_{g \in G} \models \mathbf{APr}_G^+$, then $(L_2(\mathbb{A}), \pi_g)_{g \in G} \models \mathbf{IHS}_G^+$.

In other words, the construction of the Koopman representation preserves existential closedness.

The converse does not hold.

The Koopman representation

We give a proof for $G = \mathbb{Z}$.

Example ($G = \mathbb{Z}$)

Assume $(\mathbb{A}, \tau) \models \mathbf{APr}_G^+$.

Fix a primitive n -th root of unity $\zeta \in S^1$. For $\varepsilon > 0$, let $a_\varepsilon \in \mathbb{A}$ be such that $\mu(\bigcup_{k < n} \tau^k(a_\varepsilon)) > 1 - \varepsilon$, and let $f_\varepsilon \in L_2(\mathbb{A})$ be given by

$$f_\varepsilon = \chi_{a_\varepsilon} + \zeta \chi_{\tau(a_\varepsilon)} + \zeta^2 \chi_{\tau^2(a_\varepsilon)} + \cdots + \zeta^{n-1} \chi_{\tau^{n-1}(a_\varepsilon)}.$$

Thus, we have

$$\left\| \pi f_\varepsilon - \zeta f_\varepsilon \right\| \xrightarrow{\varepsilon \rightarrow 0} 0 \quad \text{and} \quad \left\| f_\varepsilon \right\| \xrightarrow{\varepsilon \rightarrow 0} 1$$

so they witness that $\zeta \in \sigma(\pi)$.

As $\sigma(\pi)$ contains the roots of unity and it is closed, we have $\sigma(\pi) = S^1$. Thus $(L_2(\mathbb{A}), \pi) \models \mathbf{IHS}_G^+$.

Thanks!

¡Gracias por la atención!

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